Unit-2 : Data Processing Circuits.

⇒ Multiplexers:-

→ Multiplexing means transmitting a large number of information units over a smaller number of channels or lines.

→ Multiplexer is a circuit with many inputs but only one output.

→ By applying control signals, it's possible to steer any input to the output. Thus it is called a data selector and control inputs are termed as select inputs.

→ The below diagram illustrates the general idea

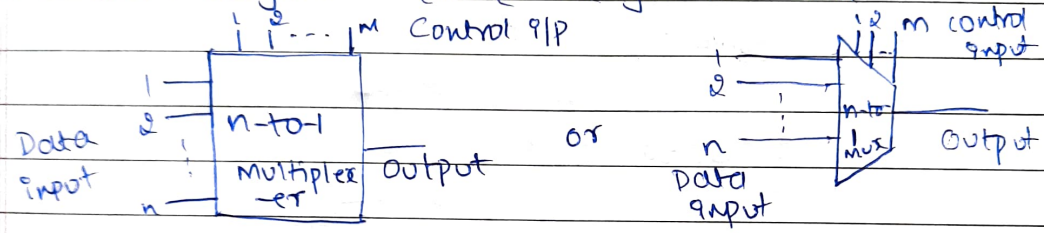


Fig. Multiplexer block diagram.

→ The circuit has n input signals, m control signals & 1 output signal. The m control signals can select at the most 2^m input signals thus $n \leq 2^m$

→ The advantage of multiplexer are

i) It reduces the number of electrical connections/wireless channels for transmission of several signals.

ii) It reduces the cost, simplifies the device electronics & provides direct interface with the microprocessors.

→ Example: The cable TV receives all the channels over a single direct interface or wire only because of multiplexing

→ The multiplexers can also be called as mux or data selector.

iii) Implementation of various circuits using mux. (Ex: adders, subtractors & so on).

→ The applications of multiplexers are

i) Communication Systems - audio or video can be sent

and received through different channels via single cable.

ii) Computer memory: to maintain a huge amount of memory in the computers & also to reduce the number of copper lines required to connect the memory to other part of the system.

iii) Telephone networks

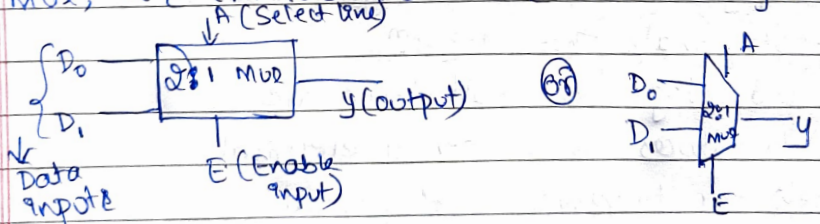
iv) Transmission from the computer system of a satellite using GSM satellite from spacecraft to ground system

→ There are few types of multiplexers:

i) 2 to 1 mux
ii) 4 to 1 mux
iii) 8 to 1 mux
iv) 16 to 1 mux
v) 32 to 1 mux

2-to-1 multiplexer:

→ The below diagrams illustrate the general idea of 2-to-1 mux, it's truth table and it's circuit diagram:



(a) Block diagram

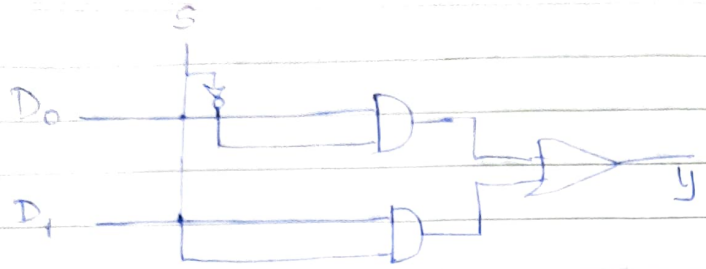
E	S	Y
0	X	0
1	0	D_0
1	1	D_1

$Y = \bar{S} \cdot E \cdot D_0 + E \cdot S \cdot D_1$

$Y = E(\bar{S}D_0 + SD_1)$

$Y = \bar{S}D_0 + SD_1$

(b) Truth table & Boolean equation.



(c) Logic diagram or circuit diagram.

→ The 2-to-1 multiplexer takes two input lines and one select lines & one output.

→ Each of the input lines, D_0, D_1 is applied to one input of an AND gate. Selection line S is decoded to select a particular AND gate.

→ The function table from the figure lists the input-to-output path for each possible bit combination of the selection lines.

→ To demonstrate the circuit operation, consider the case when S=0. The AND gate associated with D_0 input is connected together & given to the output. when S=1, the AND gate is associated with input D_1 & selected as output.

→ The another input Enable(E) basically makes the multiplexer operate.

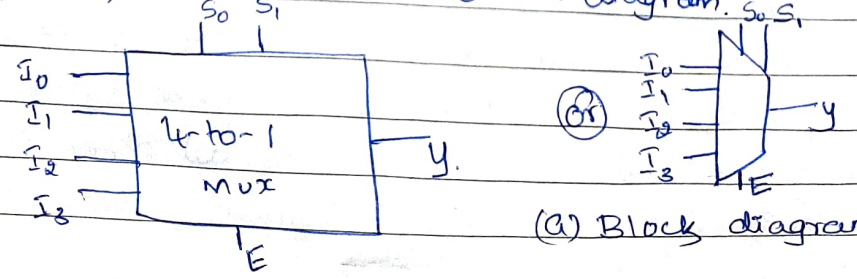
→ The Enable input is also called as strobe or tristate.

→ If the Enable input is low, it makes the multiplexer activate. Otherwise it deactivates the multiplexer.

→ Finally, the multiplexers are a medium scale integrated circuit which are available in the form of IC's.

4-to-1 Multiplexer:

∴ The below diagrams illustrates the general idea of 4-to-1 multiplexer, truth table and circuit diagram.

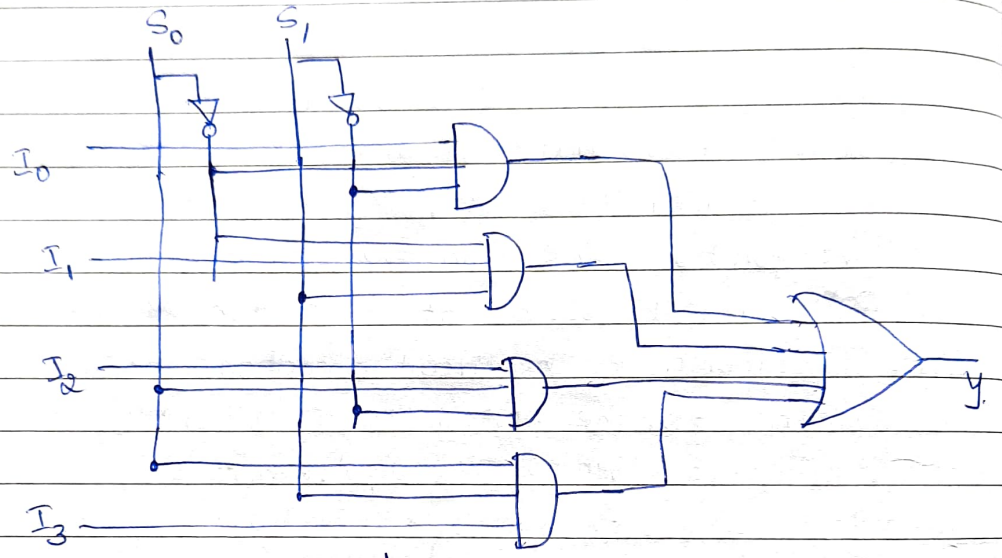


(a) Block diagram.

E	S_0	S_1	Y
0	X	X	0
1	0	0	I_0
1	0	1	I_1
1	1	0	I_2
1	1	1	I_3

$$Y = \bar{S}_0 \bar{S}_1 I_0 + \bar{S}_0 S_1 I_1 + S_0 \bar{S}_1 I_2 + S_0 S_1 I_3$$

(b) Truth table & its boolean expression



(c) circuit diagram of 4 to 1 multiplexer.

→ A 4-line to 1-line multiplexer takes 4 input, 2 data select lines & a output.

→ Each of the 4 input lines, I_0 to I_3, is applied to one input of an AND gate. Selection lines S_1 & S_0 are decoded to select a particular AND gate.

→ The above truth table (b) gives the list of the input-to-output path for each possible bit combination of the selection lines.

→ Depending on the selection lines S_0, S_1 one of the 4 inputs I_0 to I_3 is steered to output Y.

→ To demonstrate the working of circuit, consider the case for $S_0S_1 = 00$, the first AND gate to which I_0 is connected remains active and equal to D_0 and all other AND gate are inactive with output held at logic 0. Thus multiplexer

output Y is same as D_0. If $D_0=0$, $Y=0$ & if $D_0=1$ then $Y=1$.

→ Similarly, for $S_0S_1=01$, second AND gate will be active and all other AND gates remain inactive. Thus output $Y=D_1$.

→ The IC-74153 contains dual 4-to-1 MUX as shown in the below diagram.

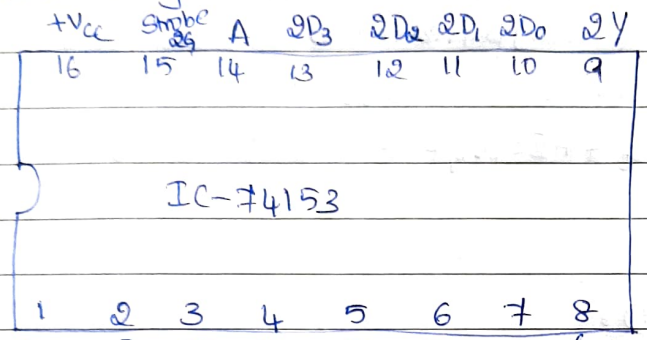


Fig. Pinout diagram for IC-74153 (4-to-1 MUX)

16-to-1 Multiplexer:

→ The below figure shows a 16-to-1 multiplexer. The input bits are labeled D_0 to D_{15}. Only one of these is transmitted to the output.

→ The control input is the dependent for the value of ABCD. For instance when

ABCD=0000

the upper AND gate is enabled while all other AND gates are disabled.

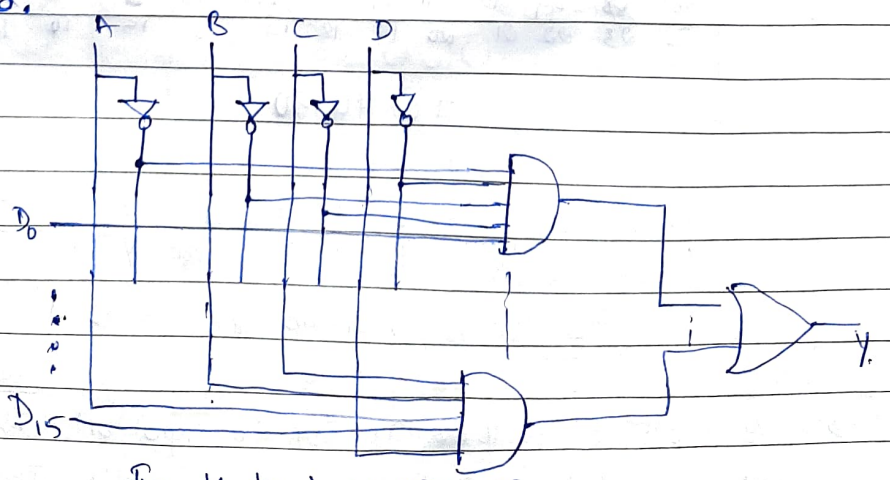


Fig. 16-to-1 Multiplexer

→ Therefore, data bit D_0 is transmitted to the output, giving

$$Y = D_0.$$

→ If D_0 is low, Y is low; if D_0 is high, Y is high. The point is that Y depends only on the value of D_0.

→ The control nibble (group of 4-bits) determines which of the input data bit is transmitted to the output.

→ Thus we can write output as

$$Y = \bar{A}\bar{B}\bar{C}\bar{D}.D_0 + \bar{A}\bar{B}\bar{C}D.D_1 + \cdots + ABCD.D_{15}.$$

The 74150:

→ Instead of using OR gate for 16 input, if we make use NOR gate, the operation of the circuit would be almost none.

→ This is because we get the complement of the selected data bit rather than the data bit itself.

→ For instance, when ABCD = 0111, the output is

$$Y = \bar{D}_7.$$

This is the boolean equation for a typical transistor-transistor logic (TTL) multiplexer because it has an inverter on the output that produces the complement of the selected data bit

→ The 74150 is a 16-to-1 TTL Multiplexer with a pin diagram as shown in the below figure:

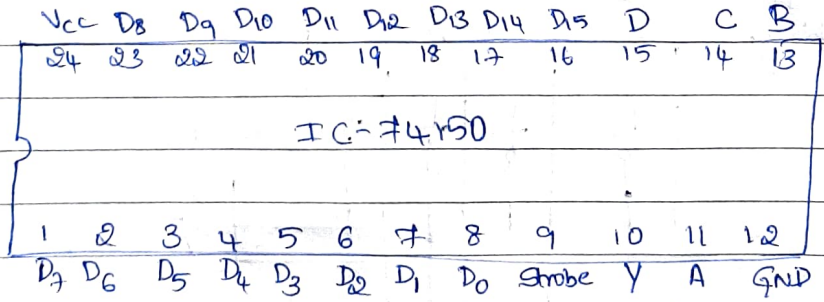


Fig. Pinout diagram of 74150

→ Pins 1 to 8 & 16 to 23 are for the input data bits D_0 to D_{15}.

→ Pins 11, 13, 14, & 15 are for the control bits ABCD.

→ Pin 10 is the output & it equals to the complement of the selected data bit.

→ Pin 9 is for the STROBE, an input signal that disables or enables the multiplexer.

→ From the below truth table

→ A low strobe enables the multiplexer, so that output, for instance Y equals the complement of the i/p data bit;

$$Y = \bar{D}_n$$

where n is the decimal equivalent of ABCD.

→ On the other hand, a high strobe disables the multiplexer and forces the output into the high state.

→ The truth table is given below:

Strobe	A	B	C	D	Y
0	0	0	0	0	$\bar{D}_0$
0	0	0	0	1	$\bar{D}_1$
0	0	0	1	0	$\bar{D}_2$
0	0	0	1	1	$\bar{D}_3$
0	0	1	0	0	$\bar{D}_4$
0	0	1	0	1	$\bar{D}_5$
0	0	1	1	0	$\bar{D}_6$
0	0	1	1	1	$\bar{D}_7$
0	1	0	0	0	$\bar{D}_8$
0	1	0	0	1	$\bar{D}_9$
0	1	0	1	0	$\bar{D}_{10}$
0	1	0	1	1	$\bar{D}_{11}$
0	1	1	0	0	$\bar{D}_{12}$
0	1	1	0	1	$\bar{D}_{13}$
0	1	1	1	0	$\bar{D}_{14}$
0	1	1	1	1	$\bar{D}_{15}$
1	X	X	X	X	1-H

Fig. Truth table of IC-74150

→ The multiplexer is active (enabled) when the strobe is low & inactive (disabled) when it is high. Because of this, the strob

⇒ Demultiplexers:-

→ Demultiplex means one into many.

→ A demultiplexer (or de-mux or data distributor) is a logic circuit which takes one single input data line & then switches it to any one of a number of individual output lines one at a time.

→ The below diagram illustrates the general idea.

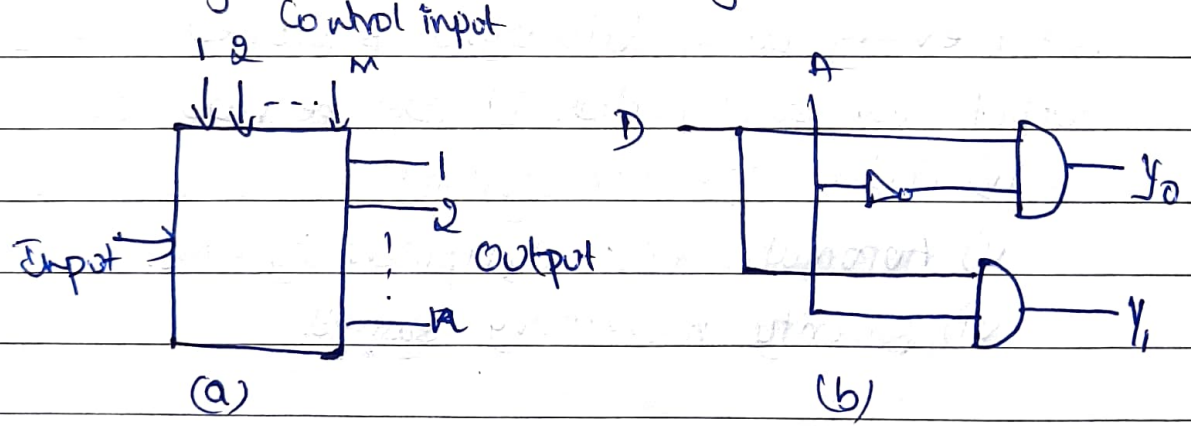


Fig. a) Demux blocks diagram
(b) Logic circuit of 1 to 2 demux
(c) Few commercially available ICs

ICNO	Demux type
74154	1-to-16
74138	1-to-8
74155	1-to-4

→ The circuit has 1 input signal, m control signal or select signals and n output signals where $n \leq 2^m$.

→ Fig.(b) shows logic circuit of 1-to-2 demultiplexer & fig(c) shows some of commercially available IC's for various demultiplexer types.

→ There are few advantages of demultiplexer that are used in microprocessor or computer control systems such as

i) selecting different IO devices for data transfer

ii) choosing different banks of memory.

iii) depends on the address, enabling different rows of memory chips.

iv) enabling different functional units

→ Few applications of demultiplexers are:

i) Communication system

Demultiplexer helps at receiving end to work simultaneously

ii) Arithmetic & Logic unit

with the implementation of demultiplexer, the output of the ALU can keep in storage units or multiple registers

iii) Serial to Parallel converter

The input of the demultiplexer is the input serial data. After every bit of data signal is stored, the demultiplexer output can be extracted. It can be read out in parallel.

iv) Data acquisition systems

v) Automatic test equipment systems.

vi) Security monitoring systems.

★ 1-to-16 Demultiplexer:-

→ 1-to-16 demultiplexer has one input data, four select lines A, B, C & D & 16 output lines Y_0 to Y_{15}. This is implemented using AND & NOT gate.

→ The input bit is labeled I. The data bit (I) is transmitted to the data bit of the output lines.

→ When ABCD = 0000, the upper AND gate is enabled while

all other AND gates are disabled, therefore, data bit I is transmitted only to the Y_0 output, giving $Y_0 = D$.

→ If D is low, Y_0 is low. If D is high, Y_0 is high.

→ Here, the value of Y_0 depends on the value of D. All other outputs are in the low state.

→ If the control nibble is changed to ABCD = 1111, all gates are disabled except the bottom AND gate. Then I, is transmitted only to the Y_{15} output, & $Y_{15} = I$

☆ The 74154:

→ The 74154 is a 1-to-16 demultiplexer with the pin. diagram is from the below figure.

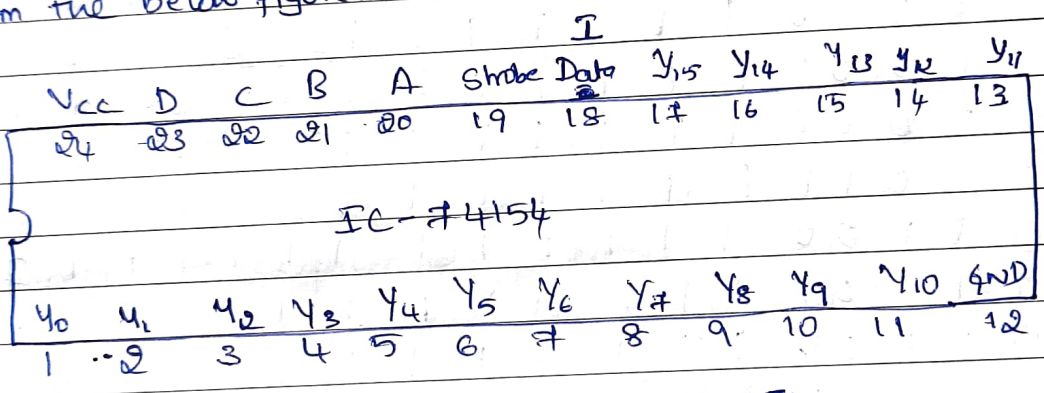


Fig. Pinout diagram of 74154

where Pin 18 – input data I.

Pins 20-23 – control bits ABCD

Pins 1 to 11 & 13 to 17 – output bits Y_0 to Y_{15}

Pin 19 – Strobe (an active-low input).

Pin 24 – V_{cc}, Pin 12 – Ground.

→ The below figure shows the figure 1-to-16 demultiplexer

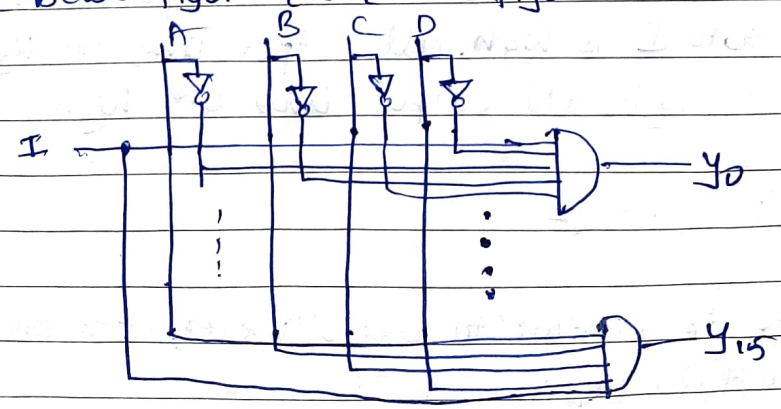


Fig: 1 to 16 demux

→ The below table shows the truth table of a 74154.

Strobe	Data I	A	B	C	D	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Y_8	Y_9	Y_{10}	Y_{11}	Y_{12}	Y_{13}	Y_{14}	Y_{15}
0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	1	0	0	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1
0	0	0	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1
0	0	0	1	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1
0	0	1	0	0	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
0	0	1	0	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1
0	0	1	0	1	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1
0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1
0	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1
0	0	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1
0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	(1) H	X	X	X	X	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	0	X	X	X	X	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

→ First, let's consider the strobe input. It must be low to activate the 74154.

→ When the strobe is low, the control input ABCD determines which output lines are low when the data I input is low.

→ When the data I is high, all output lines are high. And when the strobe is high, all output lines are high.

⇒ 1-of-16 Decoder:-

→ A decoder is similar to demultiplexer, with one exception & there is no data input.

→ The only inputs are the control input ABCD.
→ The below diagram shows the logic circuit for 1-of-16 decoder, because only 1 of the 16 output lines is high.

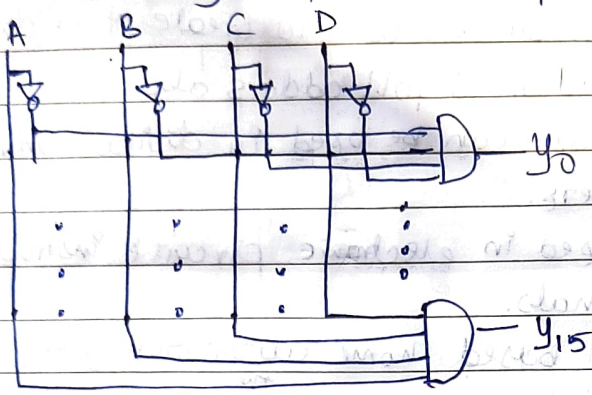


Fig. 1-of-16 decoder.

→ For instance, when ABCD is 0001, only Y_1 AND gate has all inputs high; therefore, only the Y_1 output is high.
→ The 1-of-16 decoder is also called a binary-to-decimal decoder. Because it has 4 inputs & 16 outputs, the circuit is also called as a 4-line to 16-line decoder.
→ IC 74154 is called a decoder-demultiplexer because it can be used either as a decoder or as a demultiplexer.
→ IC-74154 can be used for decoder, all you have to do is ground the Data I & Strobe inputs together. as shown in the below figure.
→ Then the selected output line is in the low state. This is why bobbles are shown on the output lines.
→ This means the output line is low when it is active or selected.
For example, if the binary input is ABCD=0111, then the Y_7 output is low, while all other outputs are high.

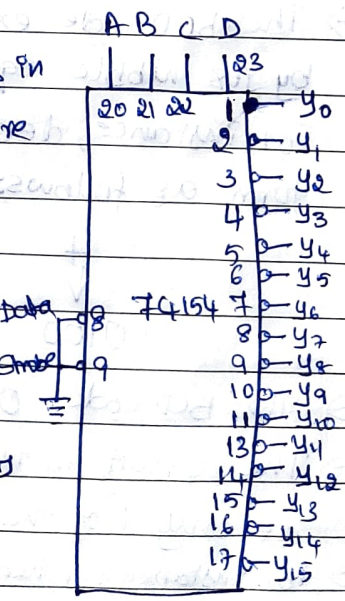


Fig. Using 74154 as decoder

→ Some of the applications of decoders are:

(1) They are mainly used as address decoders in CPU memory location identifications

(2) They can be used to create other digital logics like half adders & full adders also.

(3) They can be used in digital conversion in analog decoders.

(4) Used in electronic circuits instructions into CPU control signals.

(5) RF based home automation system.

(6) Robotic vehicle with the metal detector.

(7) Speed synchronization of multiple motors in industries

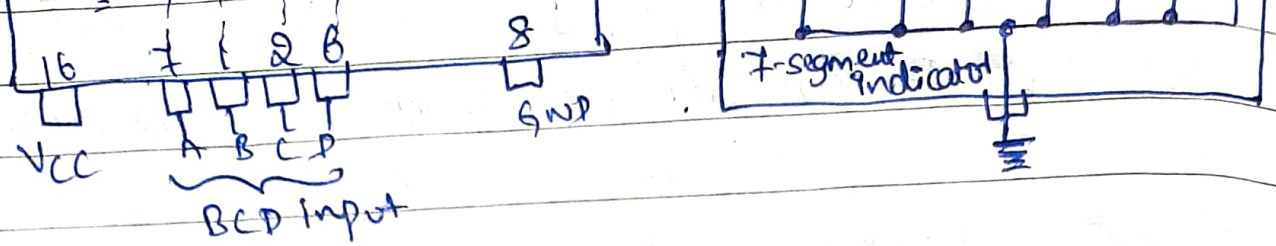


Fig. (a) 7446 decoder-driver (b) 7448 decoder-driver

✱ The 7448:

→ Fig.(b) is the alternative decoding approach. Here, a 7448 drives a common-cathode indicator.

→ Again the internal logic converts the BCD input to the required output. For example, when a BCD input of 0100 is used, the internal logic forces LEDs b, c, f & g to conduct. The seven-segment indicator then displays a 4.

→ Unlike, the 7446 that requires external current-limiting resistors, the 7448 has its own current-limiting resistors on the chip.

⇒ Encoders:-

→ An encoder is a combinational circuit that performs the reverse operation of decoder.

→ It has maximum of 2^n lines & n output lines, hence it encodes the information from 2^n inputs into an n-bit code and it will produce a binary code equivalent to the input which is active high.

→ The below figure shows the encoder

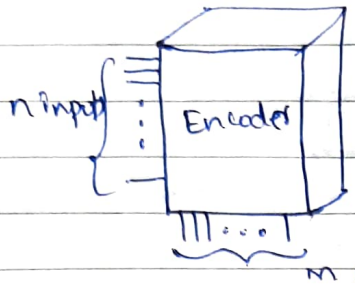


Fig. Encoder.

→ The above figure gives the illustration of how exactly a encoder looks like. with n input lines, only one of which is active. Internal logic within the encoder converts this active input to a coded binary output with m bits.

→ There are few types of encoders. They are

i) 4 to 2 Encoder
ii) 8 to 3 Encoder (octal to binary)
iii) Decimal to BCD Encoder
iv) Priority Encoder.

→ Uses of Encoders:

① Encoders are very common electronic circuits used in all digital systems.

② They are used to translate the decimal values to the binary in order to perform the binary functions such as addition, subtraction, multiplication etc.

③ Other applications especially for priority encoders may include detecting interrupts in microprocessor applications.

→ Drawbacks of Normal Encoders:

① There is an ambiguity, when all outputs of encoder are equal to zero.

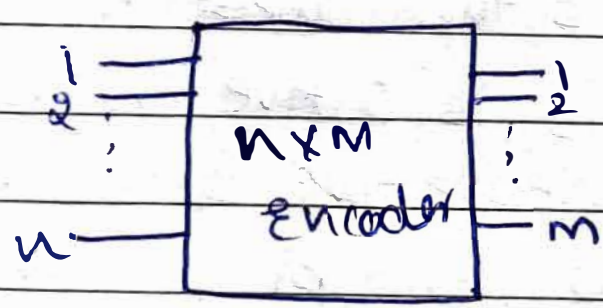
② If more than one input is active high, then the encoder produces an output, which may not be the correct code. (can be overcomed; priority encoder can be used).

Priority Encoder:-

→ A priority encoder is a circuit or algorithm that compres-ses multiple binary inputs into a smaller number of outputs.

→ If two or more inputs are given at the same time, the input having the highest priority will take precedence.

→ The below figure shows the logic symbol of priority encoder.



Here the number of input lines are indicated by 'n' & the number of output lines are indicated by 'm'.

→ The relationship between n lines and m lines is

$$n = 2^m$$

→ An example of a single bit 4 to 2 encoder is show, where highest-priority inputs are to the left and 'x' indicates an irrelevant value i.e any input value there yields the same output since it is superseded by higher-priority input. The output V indicates if the input is valid.

I_3	I_2	I_1	I_0	O_1	O_0	V
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

Truth table of 4:2 Priority encoder.

→ Some of the applications of priority encoders are:
i) Keyboard encoder
ii) Positional encoder
iii) Priority Encoder Navigation
iv) Interrupt Requests

⇒ Working of Priority Encoder:-

→ From the above truth table, it is clear that I_3, I_2, I_1 & I_0 are the inputs, O_1 & O_2 are the outputs & V is the valid bit indicator. Here I_3 input is the highest priority input & I_0 is the lowest priority input.

When the input I_3 is active high(1), which has the highest priority irrespective of all other i/p lines, then the output of the 4-bit priority encoder is 11.

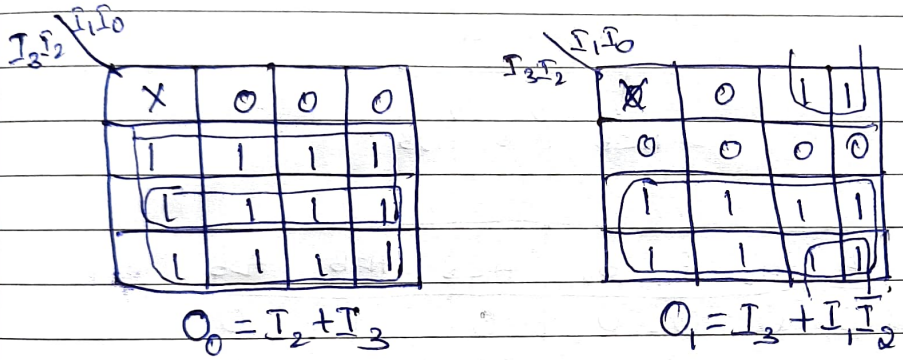
When the input I_3 is active low and the I_2 is active high that has the next highest priority irrespective of all other input lines, then the output is $O_1O_0 = 10$

When the inputs $I_3 I_2$ are active low and I_1 is active

high & has the next highest priority regardless of the remaining input line then the output will be $O_1O_0 = 01$

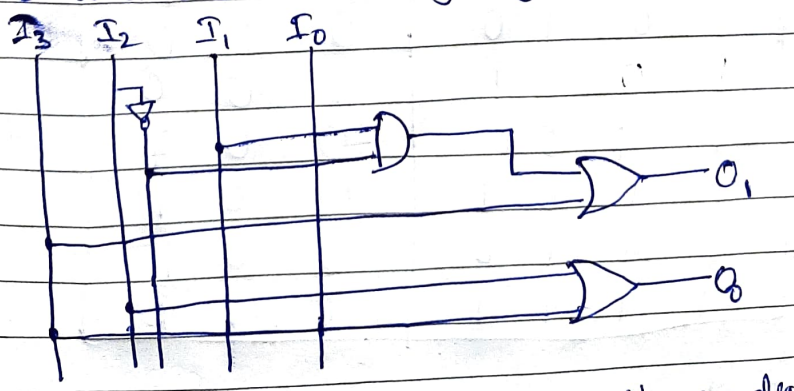
When the inputs I_3, I_2 and I_1 are active low, I_1 will be having the highest priority, then the output will be $O_1O_0 = 00$.

The output expression can be determined for a 4-bit encoder with the help of a K-map as shown below



$$O_0 = I_2 + I_3$$
$$O_1 = I_3 + I_1\bar{I}_2$$
$$V = I_0 + I_1 + \bar{I}_2 + I_3$$

From the above K-map, the simplified expressions for the outputs O_1 & O_0 are obtained. From these output expressions, the 4 to 2 priority encoder circuit diagram is illustrated with logic gates as shown below



The circuit diagram of 4 to 2 priority encoder is drawn with 2 OR gates, and the combination of AND gate and

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the NOT gate represent the valid bit, which is used when more than one input is logic high(1). Thus, 4 inputs with 2 outputs are encoded based on the assigned priority to the inputs.

⇒ Parity Generators And Checkers:-

→ The parity generating technique is one of the most widely used error detection techniques for the data transmission.

→ In digital systems, when binary data is transmitted and processed, data may be subjected to noise so that such noise can alter 0's to 1's & vice versa.

→ Hence, parity bit is added to the word containing data in order to make number of 1's either even or odd. Thus it is used to detect errors during the transmission of binary data.

→ The message containing the data bits along with parity bit is transmitted from transmitter node to receiver node.

→ At the receiving end, the number of 1's in the message is counted and if it doesn't match with the transmitting one then it means there is an error in the data.

→ A parity generator is a combinational logic circuit that generates the parity bit in the transmitter. On the other hand, a circuit that checks the parity in the receiver is called parity checker.

→ Even parity means an n-bit input has an even number of 1's.

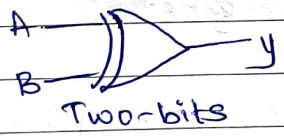
Ex:- 110011 → even parity contains 4 1's.

→ Odd parity means an n-bit input has an odd number of 1's.

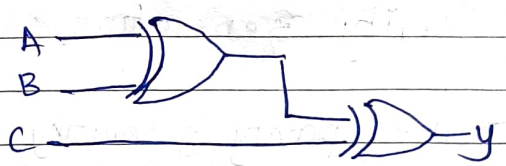
Ex:- 110001 → Odd parity contains 3 1's.

→ The basic principle involved in the implementation of parity circuits is that sum of odd number of 1's is always 1 & sum of even numbers of 1's is always 0. Such error detecting and correction can be implemented by using Ex-OR gates (since Ex-OR gate produce 0 output when there are even number of inputs).

→ To produce two bits sum, one Ex-OR gate is sufficient whereas for adding three-bits two Ex-OR gates are required as shown in the below figures.



Two-bits



3-bits

☆ Parity checker:

→ Ex-OR gates are ideal for checking the parity of a binary number because they produce an output 1 when the input has an odd number of 1's. Therefore, an even-parity input to an ex-or gate produces a low output, while an odd-parity input produces a high output.

→ For instance, the figure (next page) shows a 16-input ex-or gate. A 16-bit number drives the input.

→ The Ex-OR gate produces an output 1 because the input

has odd parity. If the 16-bit input changes to another value, the output becomes 0 for even-parity numbers & 1 for odd-parity numbers.

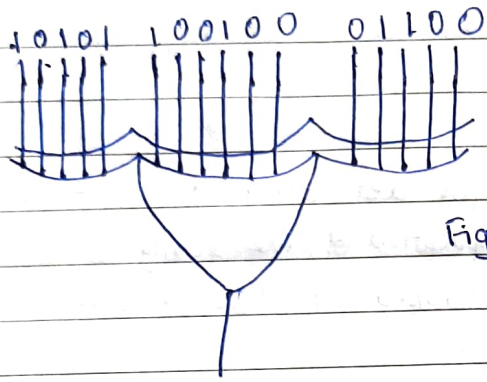


Fig. EX-OR gate with 16 inputs

☆ Parity Generation:

→ The below diagram shows the 8-bit binary number

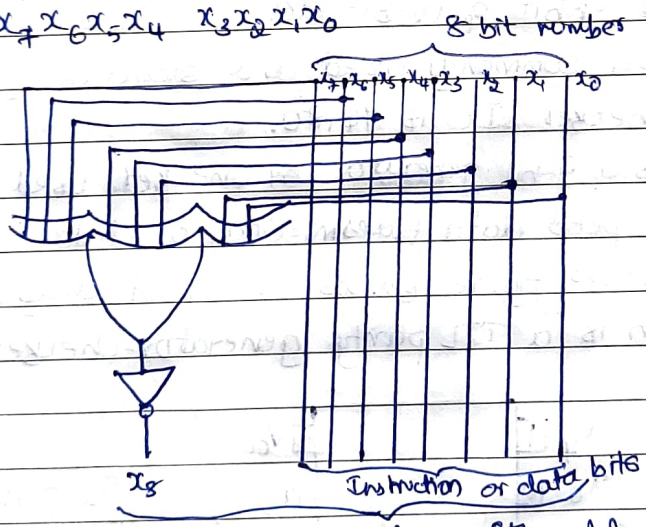


Fig. Odd parity generation.

→ Suppose this number equals 0100 0001, then the number has even parity, which means the exclusive-OR gate produces an output of 0. Because of the inverter,

$x_8 = 1$.

And the final 9-bit output is 1 0100 0001. And as we can see it has odd parity

→ Suppose we change the 8-bit input to 0110 0001. Now it

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has odd parity. In this case, the ex-OR gate produces an output. But the inverter produces a 0, so that the final 9-bit output is 0 01100001. Again the output has odd parity.

☆ Application:

→ Parity is good for detecting a single bit error.

→ One important application of the parity is to detect errors in transmitted data caused by noise or other disturbances.

⇒ Magnitude Comparator :-

⇒ Magnitude comparator is a combinational circuit that compare-s two digital or binary numbers & determines their relative magnitudes in order to find out whether one number is equal, less than or greater than the other digital number.

⇒ The below figure shows the block diagram for magnitude comparator.

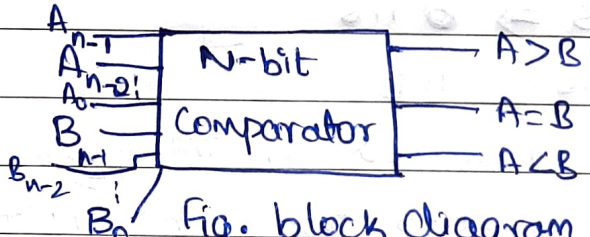


Fig. block diagram of magnitude comparator.

⇒ Few types of magnitude comparators are:

1. 1-bit comparator
2. 2-bit comparator
3. 4-bit comparator
4. 8-bit comparator

☆ 1-bit Comparator:

⇒ A comparator used to compare two bits is called a single-

bit comparator.

→ It consists of two inputs each for two single bit numbers and three outputs to generate less than, equal to & greater than between two binary numbers.

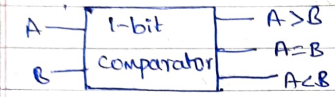
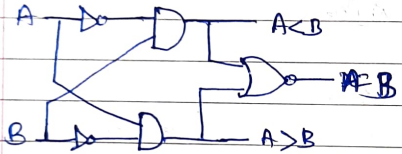


Fig. (a) block diagram of 1-bit Comparator

Inputs		Outputs		
a	b	a eq b	a gt b	a lt b
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

b) Truth table of a 1-bit Comparator



(c) Circuit for 1-Bit Comparator.

→ Fig (a) shows the block diagram of 1-bit comparator & fig.(b) gives the truth table for 1-bit comparator

→ The boolean expression for all the cases are:

$A = B \Rightarrow \overline{A}\,\overline{B} + AB = \overline{A \oplus B}$

$A > B \Rightarrow A\overline{B}$

$A < B = \overline{A}\,B$

→ The same is illustrated in the fig.(c).

→ The IC-7485 compares two 4-bit numbers as shown in the below figure:

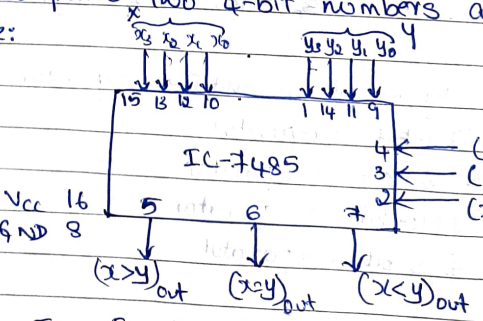


Fig. Functional diagram of IC-7485

→ This is a 16 pin IC & all the pin numbers are mentioned in the